AI and Data Science Professional

Twenty-four sessions of two hours. You arrive knowing Excel, and you leave with a model you built, tested and put online.

Bring
A laptop with a browser, a free Google account, and one question from your own work that you would like data to answer.

Module and sessions	What you do	Tools	What you make
Think with data 1 2 3 4	Read a table by its grain and keys. Chart data without misleading anyone. Simulate to see how far a sample can be trusted.	Google Sheets	A pivot table and two clear charts
Python for data 5 6 7 **8** 9	Write your first Python beside the spreadsheet formula it replaces. Load, filter and clean a messy export. Group and join tables.	Google Colab, pandas	A cleaned file that reconciles to a known total
Analyse and explain 10 11 12 **13**	Query a database with SQL. Explore a new dataset in one sitting. Test whether a difference is real and fit a regression line.	SQLite, matplotlib, statsmodels	A one-page analysis report
Machine learning 14 15 16 17 18 **19**	Split data, fit a model and beat a baseline. Judge it with a confusion matrix and catch leakage. Segment customers, forecast visits.	scikit-learn	A tested model with its baseline
Neural networks and text 20 21 22	Watch a neural network learn in your browser. Turn text into numbers and search by meaning. Test a document search.	TensorFlow Playground, sentence-transformers	A test sheet that scores the search
Ship it and present 23 24	Save a model and publish a small app. Check it for fairness and data protection. Present your capstone in ten minutes.	GitHub, Streamlit	Your app and a ten-minute talk

● Checkpoint sessions: 8, 13 and 19

Capstone. Question in session 13, dataset in 14, tested model in 19, app in 23, ten-minute talk in 24.

Every session. Recall 10 minutes, learn 30, see it done live 25, break 5, lab 40, check 10.

Everything is free. No paid tool, key or graphics card. Windows or Mac, in a browser. All course data is fictional.

KLAi Professional Learning. Tutor: Jotham Martin Wambi.